AN EXPLAINER · NO PRODUCT PITCH

How UK healthcare is organised — and where care data breaks

A short walk through the structure of the NHS in England, the standards that hold it together, and the one step where information is routinely lost.

Gonela Abhilash · educational deck · sources on the last slide

STRUCTURE

Who does what

Primary care

GP practices are **independent contractors** to the NHS, not NHS-owned. Most people's lifelong record sits in a practice's own clinical system.

Secondary care

Hospitals are **NHS trusts**. Each runs its own electronic patient record, separate from the practice's.

Commissioning

Integrated care boards plan and pay for services locally. They replaced clinical commissioning groups in July 2022.

Community & pharmacy

Community pharmacies, dentists, optometrists and community trusts each hold their own records, often with no direct link to the practice.

National services

The **NHS number**, the Spine, e-Referral Service, GP Connect and the NHS App are the shared plumbing that lets separate organisations exchange anything at all.

THE CORE DIFFICULTY

There is no single patient record

There is a practice record, a hospital record, a pharmacy record and a set of national services. What joins them is not a database — it is **correspondence**.

Practice system

EMIS Web, SystmOne and Vision hold the primary care record.

Hospital record

A trust EPR holds the admission, clinic letters and results.

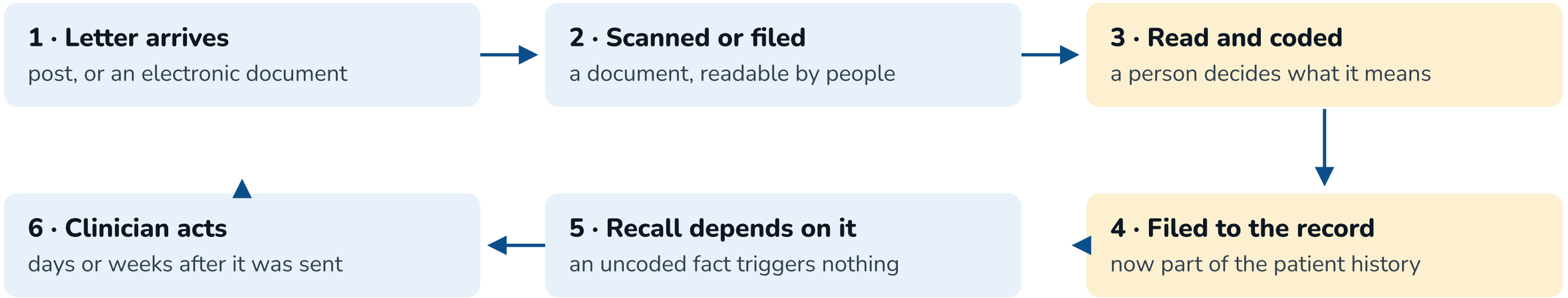
The join

Discharge summaries and clinic letters — sent to the practice as documents, often as PDFs.

Consequence: the join is manual. A human must read the letter and decide what it means for the record.

THE STEP WHERE MEANING IS LOST

One letter, six hands



Steps 3 and 4 are manual. Until they are done, the letter has arrived but the information has not. That gap is where recalls, reviews and audits quietly go wrong.

STANDARDS

What makes separate systems join up

STANDARD	WHAT IT SETTLES	WHY IT MATTERS HERE
NHS number	One identifier per person, nationally	Links a record across organisations without relying on names
SNOMED CT	The clinical terminology for direct care (standard SCCI0034)	A coded fact can be counted, recalled and audited; a paragraph cannot
dm+d	Dictionary of medicines and devices	One medicine, one code — across prescribing and dispensing
FHIR (UK Core)	How systems exchange records as data	Lets one system read another's record without bespoke work
EPS	Electronic prescription service	Prescriptions travel to a pharmacy as data, not handwriting
GP Connect / e-RS	Record sharing and referrals	Gives clinicians a route to each other's data without a letter

GOVERNANCE

The rules any record system lives under

UK GDPR & Data Protection Act 2018

Lawful basis, purpose limitation, subject access, and a DPIA before risky processing.

Caldicott principles

Eight principles on using confidential information, with a named guardian in each organisation.

DSPT

The Data Security and Protection Toolkit — the annual assurance an organisation submits about how it handles NHS data.

DTAC

The Digital Technology Assessment Criteria. Not law, but the baseline evidence base buyers ask for: clinical safety, data protection, security, usability.

DCB0129 & DCB0160

Clinical risk management: **0129** for the manufacturer of the software, **0160** for the organisation deploying it. Both need a hazard log and a named Clinical Safety Officer.

The point

None of this is optional paperwork. Each one asks the same question: can you show, not claim, how patient data is protected?

WHERE IT SHOWS UP IN A WORKING DAY

Four routine frictions

Re-reading

The same letter is read by whoever opens it — and again if it was coded wrongly.

Coding lag

Between arrival and coding, the fact exists on paper but not in the record the search engine sees.

Recall accuracy

A recall pulls patients by code. A fact recorded only in free text is invisible to the query.

Admin duplication

Data already entered in one place is typed again in another, because the systems do not speak.

These are structural, not personal. They follow from having separate systems joined by documents.

WHERE A TOOL CAN HELP

Turning a document into a coded, confirmed fact

1 · Read

Take the letter or PDF and extract the diagnosis, the medicine, the action and the dates.

2 · Propose

Suggest a SNOMED CT code for each fact, with a confidence attached, not a decision.

3 · Confirm

A clinician accepts, edits or rejects. Nothing enters the record before that.

4 · File & export

Write FHIR R4 records, linked on the NHS number, with an audit trail of who did what.

This is the design of **HealthBridge**, the project this explainer comes from. The claim is narrow: shorten the manual middle of the letter's journey, and leave the clinical decision with the clinician.

HONEST POSITION

What a UK deployment would require

- **DTAC evidence** — clinical safety case, DPIA, security documentation, usability testing
- **DSPT** — submitted and current, or the buyer will stop at the first question
- **DCB0129 hazard log** and a named Clinical Safety Officer before any pilot
- **Cyber Essentials** or ISO 27001 as the security baseline

HealthBridge holds none of these today.

It was built India-first, against DPDP 2025 and ABDM, and its code is European-readable: FHIR R4 records, tenant-scoped access, an audit log. The engineering travels. The certification does not — yet.

Saying this first is not modesty. In UK health IT, a supplier who overstates compliance is the one buyers stop calling.

SOURCES

Where to read this properly

SNOMED CT standard SCCI0034
digital.nhs.uk — information standards

Terminology releases (SNOMED, dm+d)
isd.digital.nhs.uk/trud

NHS England structure & ICBs
england.nhs.uk

The Caldicott Principles
gov.uk/government/publications/the-caldicott-principles

DSPT and DTAC
digital.nhs.uk — data security and assessment criteria

Clinical risk management DCB0129 / DCB0160
england.nhs.uk — clinical safety standards

This deck is an explainer written by **Gonela Abhilash** (CPC · registered pharmacist · MSc Public Health, Anglia Ruskin). It is not affiliated with the NHS and makes no compliance claim. Figures and structures described here may change — check the sources above for the current position.